

THANUJ RAJA

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Computer Science graduate specializing in Artificial Intelligence with strong foundations in Python, Java, SQL, Machine Learning, and Deep Learning. Experienced in developing AI-powered applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Computer Vision. Passionate about building scalable, data-driven solutions to solve real-world problems.

Education

Amrita Vishwa Vidyapeetham

B.Tech, CSE – AI (CGPA: 7.97)

Bengaluru

Sept. 2022 – 2026

Narayana Junior College

Class 12th – 88.8%, MPC

Hyderabad

Sept. 2020 – May 2022

Dr. KKR Gowtham Concept School

Class 10th – 100%, SSC

Hyderabad

Sept. 2019 – Jun 2020

Projects

RAG-Based AI Candidate Screening System

2026

- Built an end-to-end AI interview platform using Retrieval-Augmented Generation (RAG), LLMs, and semantic search, with a role-specific knowledge base from PDF extraction, vector embeddings (Sentence Transformers, all-MiniLM-L6-v2), and ChromaDB indexing for contextual retrieval.
- Implemented resume parsing, adaptive question generation, automated answer evaluation, and structured interview report generation with PostgreSQL for personalized, grounded technical assessments.

Solar Panel Fault Detection: Comparative Analysis of Deep Learning Models

2025

- Designed and compared custom CNNs, CBAM attention, and hybrid ResNet50-VGG16 models to detect faults in solar panels from aerial images.
- Applied advanced data augmentation, weighted sampling, and rigorous evaluation to achieve 97.12% classification accuracy for scalable, real-time renewable energy asset monitoring using Python and PyTorch.

Coral Reef Health Monitoring Using an Ensemble Deep Learning Model

2024

- Developed an ensemble model integrating a custom CNN (convolutional, pooling, and CBAM attention modules) with a VGG16 network, both trained from scratch and combined via majority voting.
- Achieved 84% test accuracy on large image datasets, optimizing the architecture for robust feature extraction and improved per-class classification.

Skills

- Languages:** Python, Java
- Databases:** PostgreSQL, MySQL, ChromaDB
- Frameworks/Libraries:** Pandas, PyTorch, OpenCV, Transformers, Sentence Transformers, LangChain, NLTK, Matplotlib
- Concepts:** Machine Learning, Deep Learning, LLMs, RAG, SQL, DBMS, Data Structures, OOP
- Tools/Platforms:** Git & GitHub, Visual Studio, AWS, MySQL Workbench, Hugging Face
- Interpersonal:** Project Planning, Adaptability, Leadership, Teamwork, Problem Solving, Communication

Achievements and Research

Social Media Sentiment Analysis Using Twitter Dataset

2024

- Published in the 3rd International Conference on Optimization Techniques in the Field of Engineering (ICOFE-2024).
- Built a real-time sentiment classification system evaluating Logistic Regression, Naive Bayes, Random Forest, and SVM for optimal accuracy.

IEEE-CIS Hackathon

2024

- Secured 3rd place in an overnight hackathon at Amrita Vishwa Vidyapeetham, Bengaluru.

Certifications

- IBM Skill Boost (2024): Artificial Intelligence Fundamentals
- AWS Academy Graduate (2024): Introduction to Cloud Semester 1
- MathWorks Certification (2025): MATLAB and Image Processing Onramp